

GURU TEGH BAHADUR PUBLIC SCHOOL, DURGAPUR

SESSION : 2026-27

PRE-BOARD EXAMINATION – I | MARKING SCHEME

CLASS : XII

MAXIMUM MARKS : 50

SUBJECT : INFORMATICS PRACTICES (065)

TIME ALLOWED : 1 Hr. 45 Min.

Chapters covered: Ch 1 Python Pandas – I, Ch 2 Python Pandas – II, Ch 9 Introduction to Computer Networks, Ch 10 Introduction to Internet and Web.

Marks by chapter: Pandas (Ch 1 & 2) – 30; Networks, Internet & Web (Ch 9 & 10) – 20.

SECTION - A (1 × 13 = 13)

- (a) False (b) (B) 90 (c) (C) size (d) (A) `pd.read_csv()` (e) (B) `index=False`
(f) columns (g) (A) PAN (h) (D) VoIP (i) (C) Repeater
(j) (D) Interlinking of collection of webpages is called Internet. (k) (D) Mesh
(l) (A) Both A and R are True and R is the correct explanation for A.
(m) (A) Both A and R are True and R is the correct explanation for A.

SECTION - B

- Q.2** The four blanks are `pandas`, `pd`, `di` and `'Sariska'`. (½ mark each) (2)
- Q.3** A web server is a computer or program that stores websites and sends web pages to clients when they request them. A web browser is client software used to request, retrieve and display web pages. (1)
Popular browsers: Google Chrome, Mozilla Firefox, Microsoft Edge, Safari. Any two, ½ mark each. (2)
- Q.4** (i) Index: Q1, Q2, Q3, Q4, A, B, C (1) (2)
(ii) Columns: 1, 2 (1)

SECTION - C

- Q.5** (a) `House = pd.Series({'Beas': 18, 'Chenab': 2, 'Ravi': 20, 'Satluj': 18})` (3)
(b) `print(House[House > 15])`

Output of (b), for reference:

```
Beas      18
Ravi      20
Satluj    18
dtype: int64
```

- Q.6** (a) A web page is a single document on the WWW. A website is a collection of related web pages under one domain name. (3)
(b) A static web page shows the same, fixed content to every user, and its content changes only when the page is edited. A dynamic web page's content is generated when it is requested, for example by server-side scripts or a database, so it can change for each user or visit.
(c) A hub sends incoming data to all connected devices. A switch sends data only to the intended device. (1 mark each)
- Q.7** (a) (3)
- ```
S1 166
S2 137
S3 185
S4 95
dtype: int64
```
- (b) S4 Dev  
(c) 95 55  
(1 mark each)

### **SECTION - D**

- Q.8** (a) `L = [['Divya', 'HR', 95000], ['Mamta', 'Marketing', 97000],`  
`['Payal', 'IT', 98000], ['Deepak', 'Sales', 79000]]`  
`Emp = pd.DataFrame(L, columns=['Name', 'Department', 'Salary'])`  
(b) `Emp.to_csv("EMP.csv", index=False)`  
(c) `Emp2 = pd.read_csv("EMP.csv")` (4)
- Q.9** (a) URL – Uniform Resource Locator; WWW – World Wide Web. ( $\frac{1}{2} + \frac{1}{2}$ ) (4)  
(b) Web hosting is a service that stores a website's files on a web server connected to the Internet, so that the website can be accessed by users all over the world. (1)  
(c) Add-ons or plug-ins are small software components added to a web browser to give it extra features. Examples: an ad-blocker, a PDF viewer, or a video player plug-in. ( $\frac{1}{2} + \frac{1}{2}$ )  
(d) Modem (Modulator–Demodulator). It is needed because computers work with digital signals, while telephone and cable lines carry analog signals. ( $\frac{1}{2} + \frac{1}{2}$ )
- Q.10** (a) `Genre['Num_Copies'] = [300, 290, 450, 760]` (4)  
(b) `Genre.loc[4] = ['Folk Tale', 'FT', 600]`  
(c) `Genre = Genre.drop(2)` # or `Genre.drop(2, inplace=True)`  
(d) `Genre.loc[3, 'Num_Copies'] = 500` # or `Genre.at[3, 'Num_Copies'] = 500`  
(1 mark each)

### **SECTION - E**

- Q.11** (a) 15 (5)  
(b) `df['Total Sales'] = df['Qtr1'] + df['Qtr2'] + df['Qtr3'] + df['Qtr4']`  
(c) `print(df[['Qtr1', 'Qtr2', 'Qtr3', 'Qtr4']].max(axis=1))`  
(d) `df = df.drop(2)` # or `df.drop(2, inplace=True)`  
(e) `df.to_csv("D:\\data.csv")` # or `df.to_csv("D:/data.csv")`  
(1 mark each)
- Q.12** (a) PSYCHIATRY block. It has the most computers (40), so placing the server there keeps most traffic local. (1) (5)  
(b) Switch (or hub). (1)  
(c) Star layout centred on PSYCHIATRY: PSYCHIATRY–ADMIN (65 m), PSYCHIATRY–ICU (50 m), PSYCHIATRY–PSYCHOLOGY (100 m).  
Also accept the minimum-cable layout PSYCHIATRY–ICU (50 m), ICU–PSYCHOLOGY (50 m), ADMIN–ICU (65 m). (1)  
(d) In the PSYCHIATRY block, the server block. (1)  
(e) LAN. (1)

### **SOURCES OF QUESTIONS TAKEN FROM PREVIOUS PAPERS**

| Question                 | Source                                                                                                                                                                |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1 (g)                   | CBSE CS Sample Paper 2023-24, Q6 (verbatim)                                                                                                                           |
| Q1 (h)                   | CBSE Board Exam 2025 (CS, Q.P. Code 91), Q18                                                                                                                          |
| Q1 (i)                   | CBSE Board Exam 2025 (CS, Q.P. Code 91), Q17 (verbatim)                                                                                                               |
| Q1 (j)                   | CBSE CS Additional Practice Paper 2023-24, Q15 (verbatim)                                                                                                             |
| Q3                       | CBSE CS Sample Paper 2020-21, Q25 (OR part); also CS Practice Paper 2023-24, Q19                                                                                      |
| Q12                      | CBSE Board Exam 2025 (CS, Q.P. Code 91), Q37. Part (e) is the OR option of (v).                                                                                       |
| Q2, Q4, Q5, Q8, Q10, Q11 | Adapted from CBSE IP Sample Papers (2022-23 / 2023-24). These are from memory, because no IP papers are in the repository, so check them against the original papers. |

\*\*\*\*\* End of Marking Scheme \*\*\*\*\*